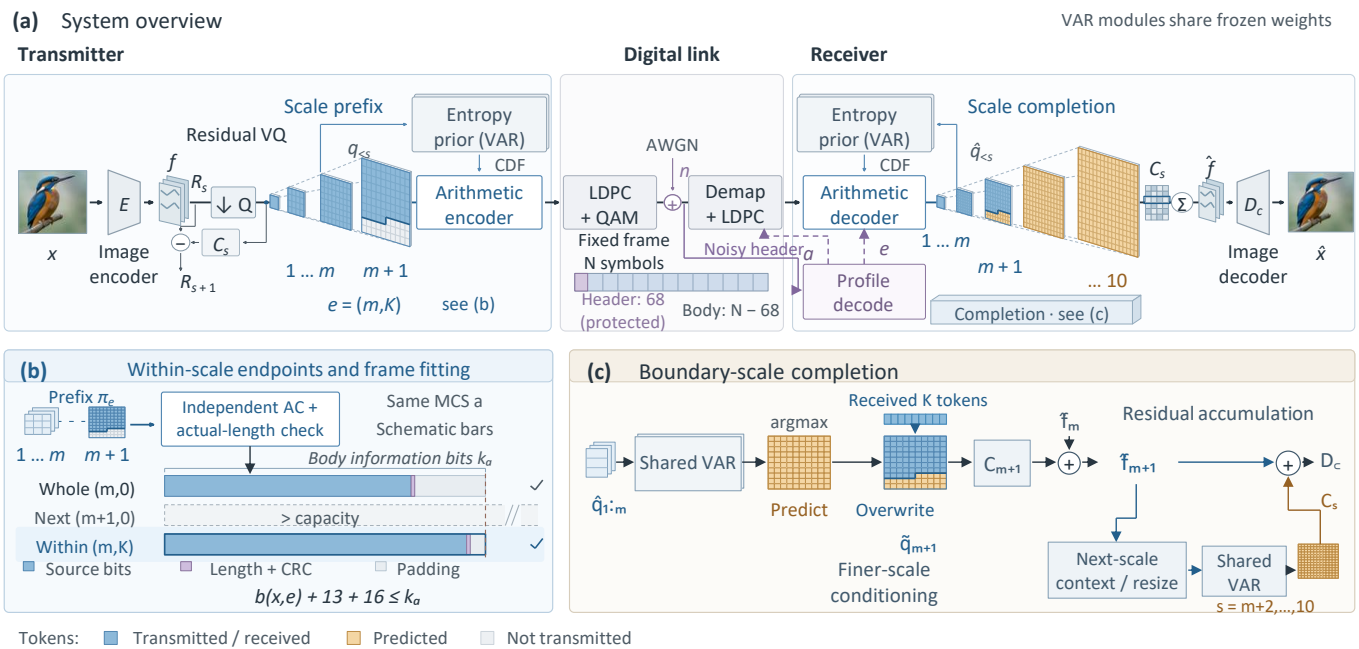

(a) System overview
VAR modules share frozen weights
Transmitter
Digital link
Receiver
Scale prefix
Residual VQ
Entropy prior (VAR)
CDF
Arithmetic encoder
Image encoder
x
E
f
R_s
↓ Q
C_s
R_{s+1}
q_{<s}
1 … m
m + 1
e = (m,K)
see (b)
AWGN
n
LDPC + QAM
Fixed frame N symbols
Demap + LDPC
Noisy header
a
Header: 68 (protected)
Body: N − 68
Entropy prior (VAR)
CDF
Arithmetic decoder
Profile decode
e
Scale completion
q̂_{<s}
1 … m
m + 1
… 10
Completion · see (c)
C_s
Σ
f̂
D_c
Image decoder
x̂
(b) Within-scale endpoints and frame fitting
Prefix π_e
1 … m
m + 1
Independent AC + actual-length check
Same MCS a
Schematic bars
Body information bits k_a
Whole (m,0)
Next (m+1,0)
> capacity
Within (m,K)
Source bits
Length + CRC
Padding
b(x,e) + 13 + 16 ≤ k_a
(c) Boundary-scale completion
Received K tokens
argmax
Predict
Overwrite
Shared VAR
q̂_{1:m}
C_{m+1}
f_m
f_{m+1}
Residual accumulation
D_c
C_s
q̃_{m+1}
Finer-scale conditioning
Next-scale context / resize
Shared VAR
s = m+2,…,10
Tokens:
Transmitted / received
Predicted
Not transmitted